Ex.No 7 **Create simple application like wordcount using hadoop**

Aim

To create simple application like word count using hadoop.

Softwares Required

- hadoop
- eclipse
- hadoop jar files

Word count

Hadoop MapReduce is a software framework for easily writing applications which process vast amounts of data (multi-terabyte data-sets) in-parallel on large clusters (thousands of nodes) of commodity hardware in a reliable, fault-tolerant manner.

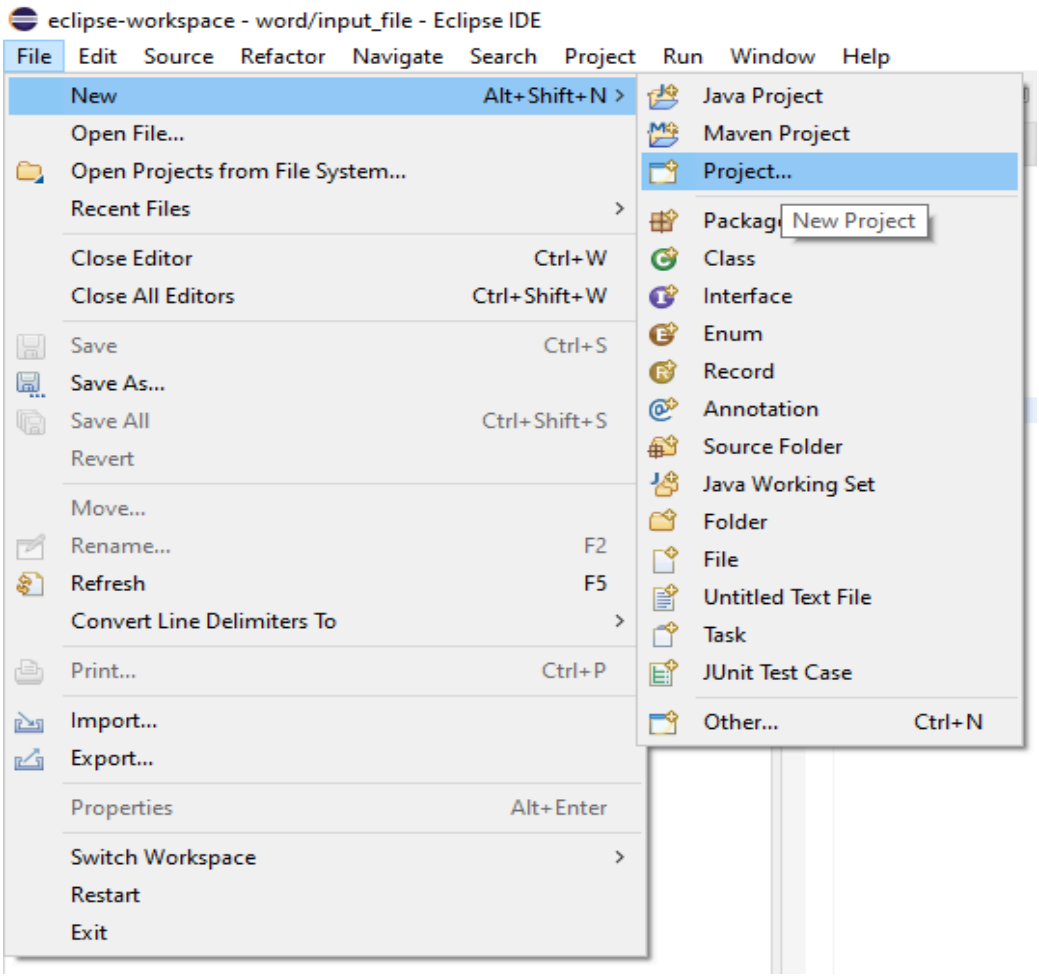
A MapReduce job usually splits the input data-set into independent chunks which are processed by the map tasks in a completely parallel manner. The framework sorts the outputs of the maps, which are then input to the reduce tasks. Typically both the input and the output of the job are stored in a file-system. The framework takes care of scheduling tasks, monitoring them and re-executes the failed tasks.

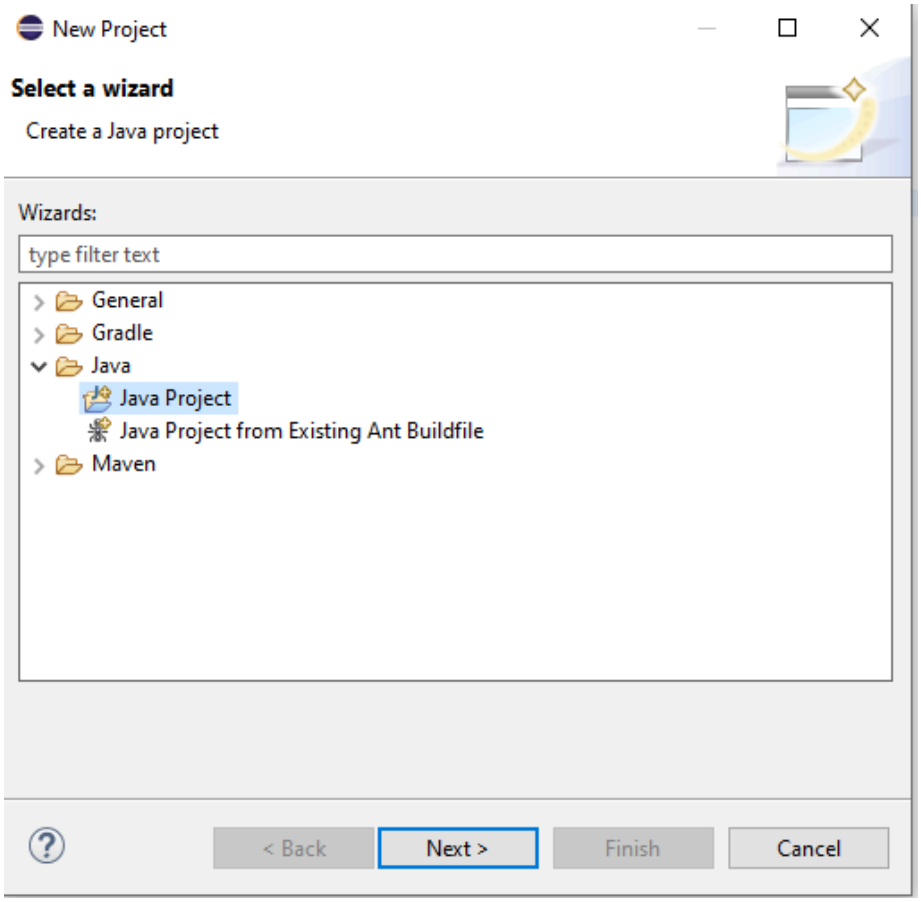
WordCount is a simple application that counts the number of occurrences of each word in a given input set.

Program

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class wordcount {

  public static class TokenizerMapper
      extends Mapper<Object, Text, Text, IntWritable>{

    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();

    public void map(Object key, Text value, Context context
```

```
                ) throws IOException, InterruptedException {
    StringTokenizer itr = new StringTokenizer(value.toString());
    while (itr.hasMoreTokens()) {
      word.set(itr.nextToken());
      context.write(word, one);
    }
  }
}

public static class IntSumReducer
     extends Reducer<Text,IntWritable,Text,IntWritable> {
  private IntWritable result = new IntWritable();

  public void reduce(Text key, Iterable<IntWritable> values,
                     Context context
                     ) throws IOException, InterruptedException {
    int sum = 0;
    for (IntWritable val : values) {
      sum += val.get();
    }
    result.set(sum);
    context.write(key, result);
  }
}

public static void main(String[] args) throws Exception {
  Configuration conf = new Configuration();
  Job job = Job.getInstance(conf, "word count");
  job.setJarByClass(wordcount.class);
  job.setMapperClass(TokenizerMapper.class);
  job.setCombinerClass(IntSumReducer.class);
  job.setReducerClass(IntSumReducer.class);
  job.setOutputKeyClass(Text.class);
  job.setOutputValueClass(IntWritable.class);
  FileInputFormat.addInputPath(job, new Path(args[0]));
  FileOutputFormat.setOutputPath(job, new Path(args[1]));
  System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```


eclipse-workspace - word/input_file - Eclipse IDE
File
Edit
Source
Refactor
Navigate
Search
Project
Run
Window
Help
New Alt+Shift+N
Open File...
Open Projects from File System...
Recent Files
Close Editor Ctrl+W
Close All Editors Ctrl+Shift+W
Save Ctrl+S
Save As...
Save All Ctrl+Shift+S
Revert
Move...
Rename... F2
Refresh F5
Convert Line Delimiters To
Print... Ctrl+P
Import...
Export...
Properties Alt+Enter
Switch Workspace
Restart
Exit
Java Project
Maven Project
Project...
Packag
New Project
Class
Interface
Enum
Record
Annotation
Source Folder
Java Working Set
Folder
File
Untitled Text File
Task
JUnit Test Case
Other... Ctrl+N


New Project
Select a wizard
Create a Java project
Wizards:
type filter text
General
Gradle
Java
Java Project
Java Project from Existing Ant Buildfile
Maven
< Back
Next >
Finish
Cancel

New Java Project

Create a Java Project

Create a Java project in the workspace or in an external location.

Project name: ex8cloud

☑ Use default location

Location: C:\Users\DCE\eclipse-workspace\ex8cloud Browse...

JRE

◉ Use an execution environment JRE: JavaSE-17

○ Use a project specific JRE: jre

○ Use default JRE 'jre' and workspace compiler preferences Configure JREs...

Project layout

○ Use project folder as root for sources and class files

◉ Create separate folders for sources and class files Configure default...

Working sets

☐ Add project to working sets New...

Working sets: Select...

Module

☑ Create module-info.java file

Module name:

☑ Generate comments

module name will be "ex8cloud" (if no module is specified, then project name will be used as module name)

< Back Next > Finish Cancel

eclipse-workspace - Eclipse IDE

File Edit Source Refactor Navigate Search Project

Package Explorer

- ex8cloud
  - JRE System Library [JavaSE-17]
  - src

New Java Class

Java Class

Type name is discouraged. By convention, Java type names usually start with an uppercase letter

Source folder: ex8cloud/src Browse...

Package: ex8cloud Browse...

Enclosing type: Browse...

Name: wordcount

Modifiers: public package private protected

abstract final static

none sealed non-sealed final

Superclass: java.lang.Object Browse...

Interfaces: Add... Remove

Which method stubs would you like to create?

public static void main(String[] args)

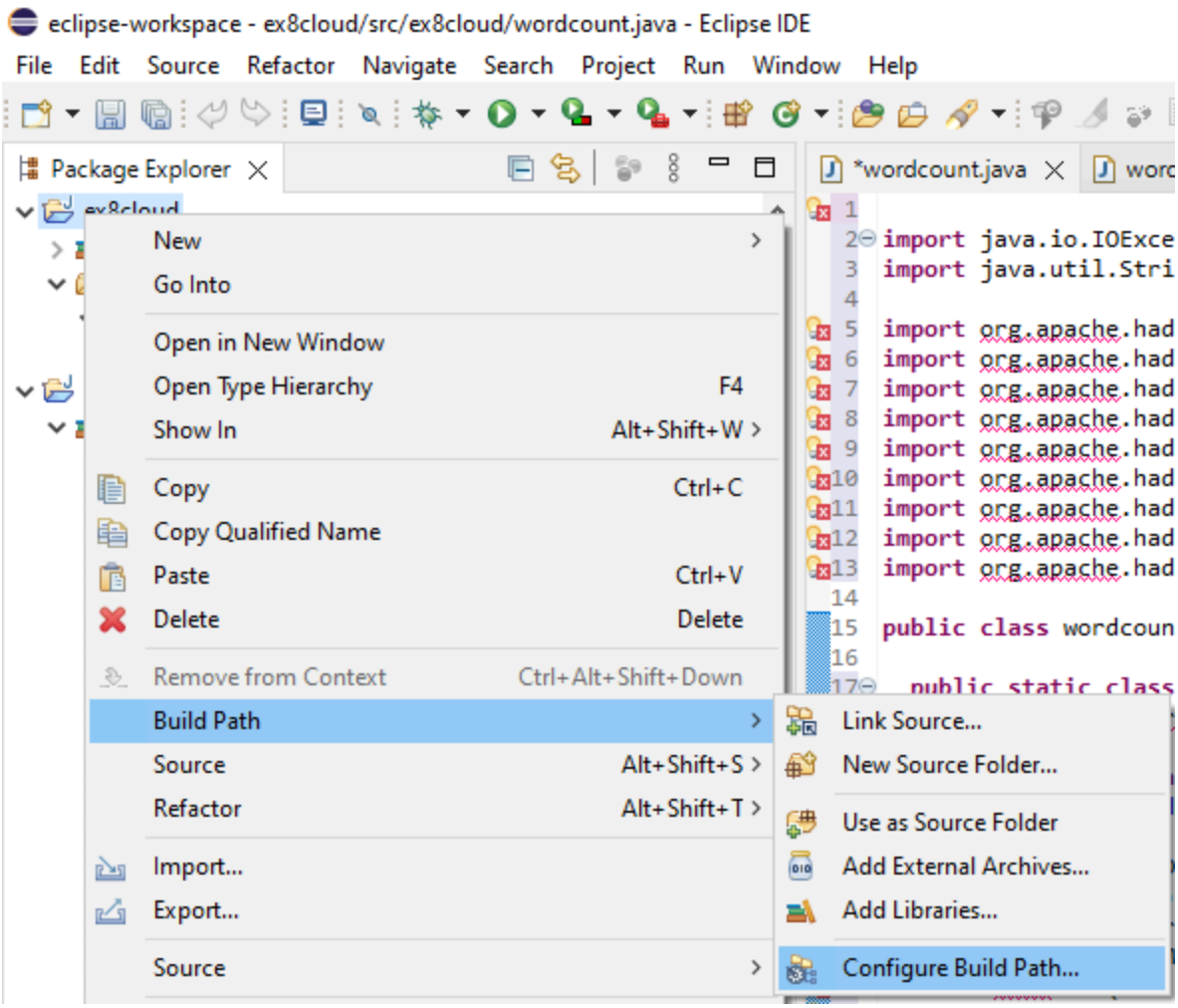
Constructors from superclass

Inherited abstract methods

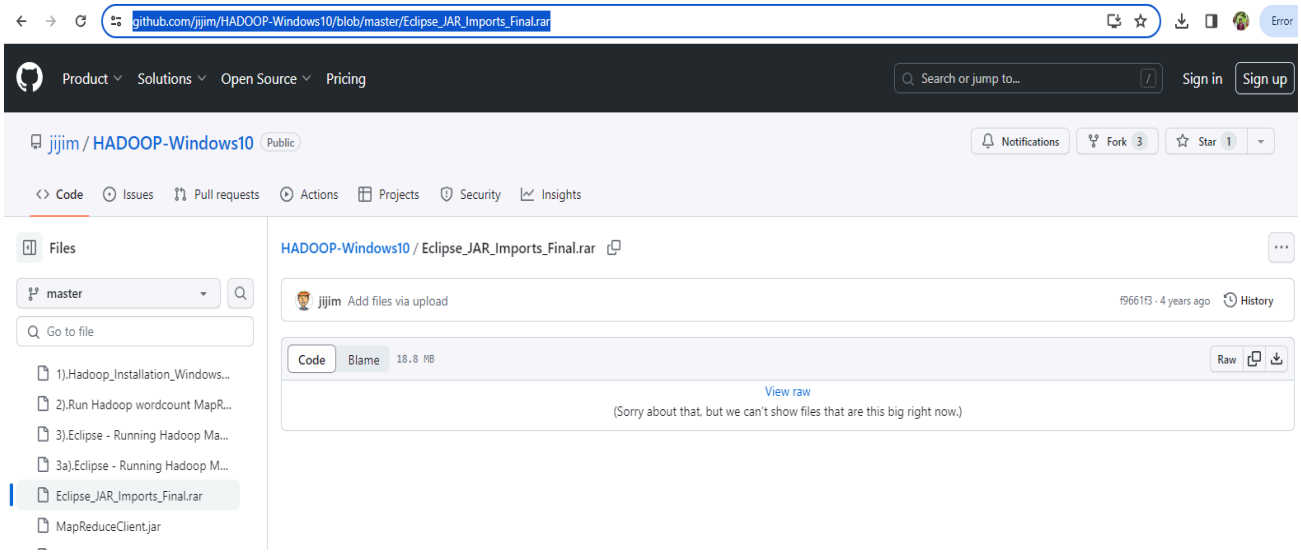
Do you want to add comments? (Configure templates and default value here)

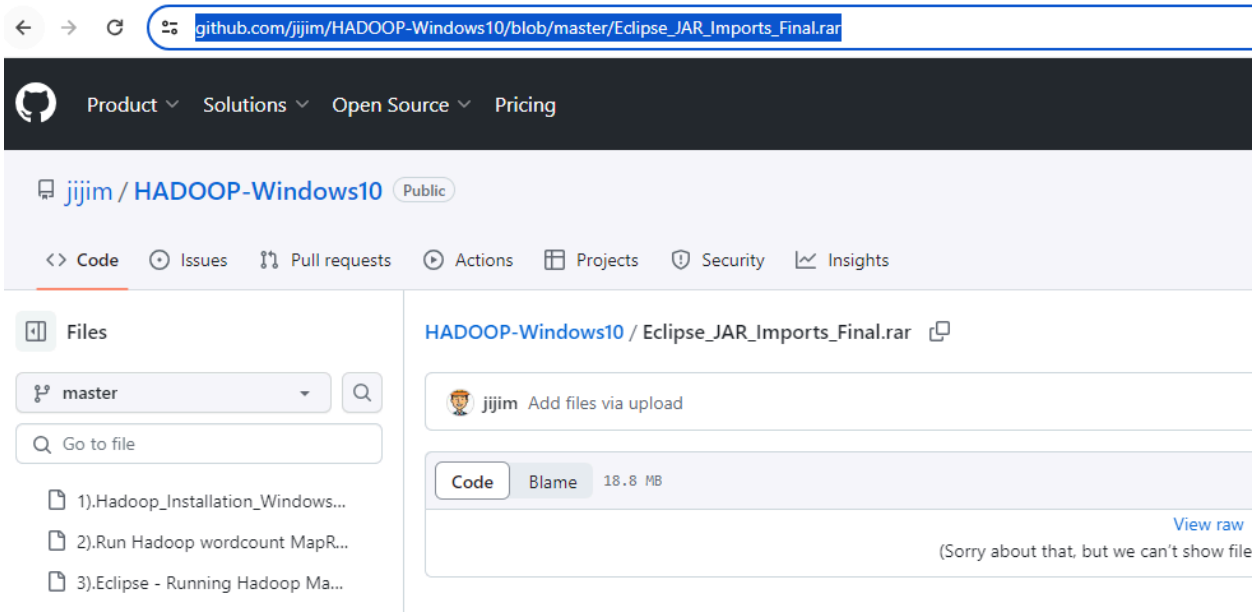
Generate comments

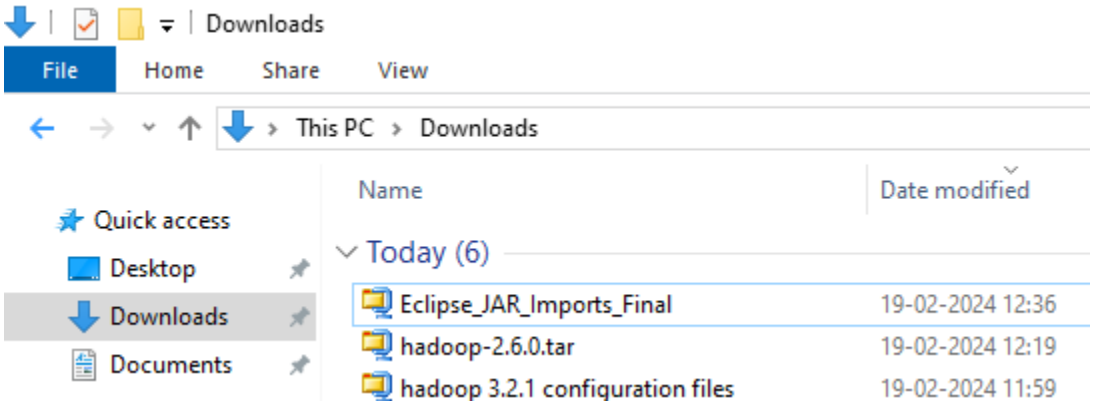
Finish Cancel



https://github.com/jijim/HADOOP-Windows10/blob/master/Eclipse_JAR_Imports_Final.rar




github.com/jijim/HADOOP-Windows10/blob/master/Eclipse_JAR_Imports_Final.rar
Product
Solutions
Open Source
Pricing
jijim / HADOOP-Windows10
Public
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Files
master
Go to file
1).Hadoop_Installation_Windows...
2).Run Hadoop wordcount MapR...
3).Eclipse - Running Hadoop Ma...
HADOOP-Windows10 / Eclipse_JAR_Imports_Final.rar
jijim Add files via upload
Code
Blame
18.8 MB
View raw
(Sorry about that, but we can't show file


Downloads
File
Home
Share
View
This PC
Downloads
Quick access
Desktop
Downloads
Documents
Name
Date modified
Today (6)
Eclipse_JAR_Imports_Final
19-02-2024 12:36
hadoop-2.6.0.tar
19-02-2024 12:19
hadoop 3.2.1 configuration files
19-02-2024 11:59

Eclipse_JAR_Imports_Final

File Home Share View

This PC › New Volume (F:) › Eclipse_JAR_Imports_Final

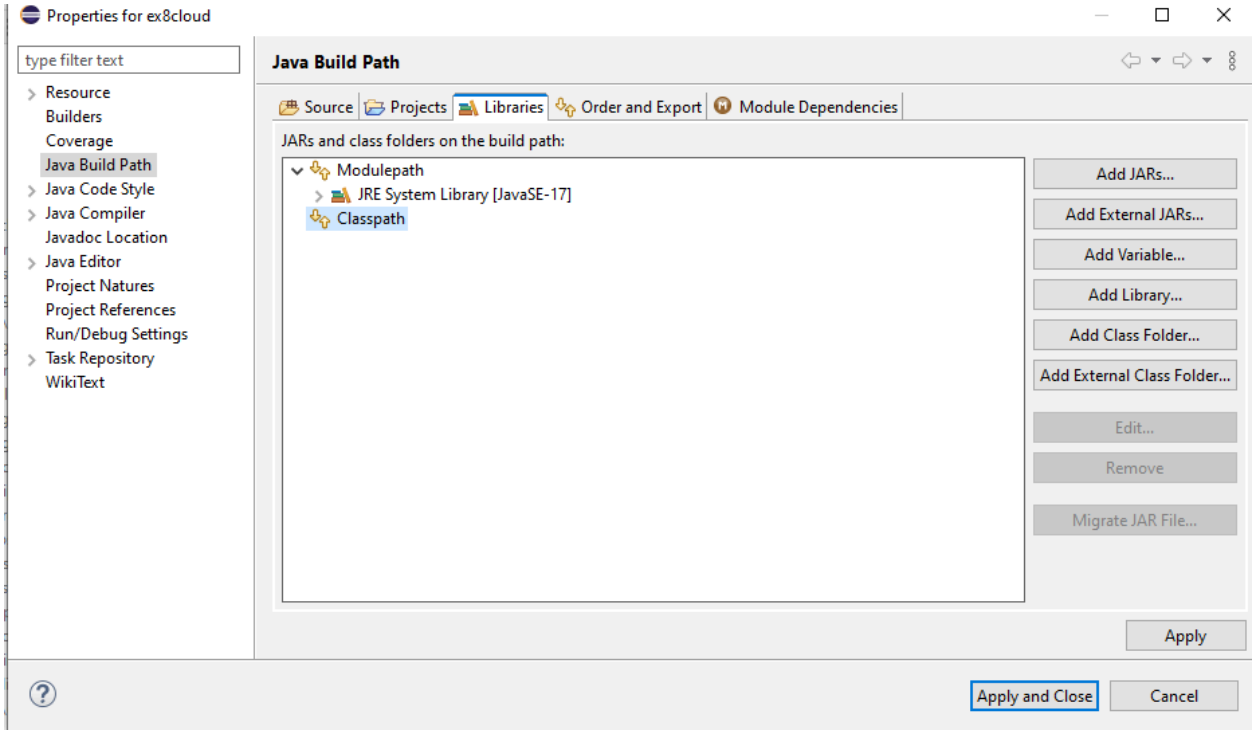
Quick access
- Desktop
- Downloads
- Documents
- Pictures
- hadoop
- hadoop
- hadoop-3.3.0
- New Volume (F:)

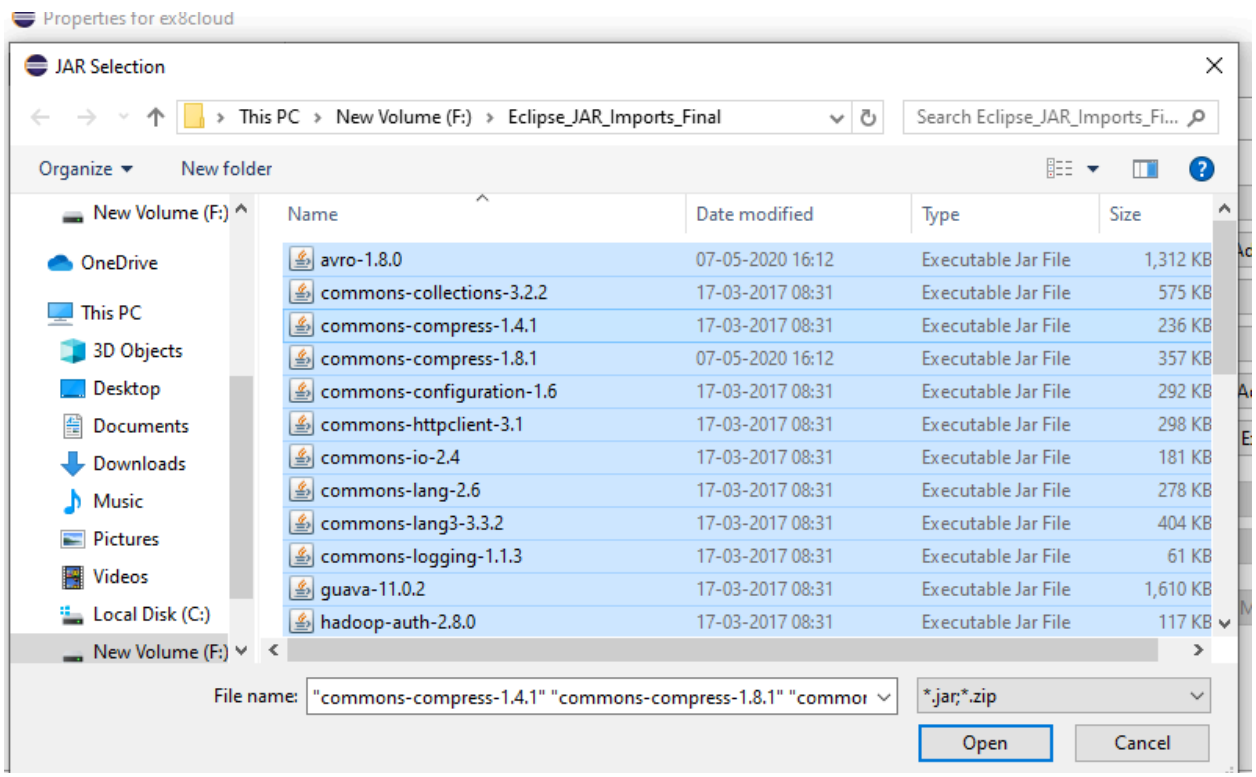
OneDrive

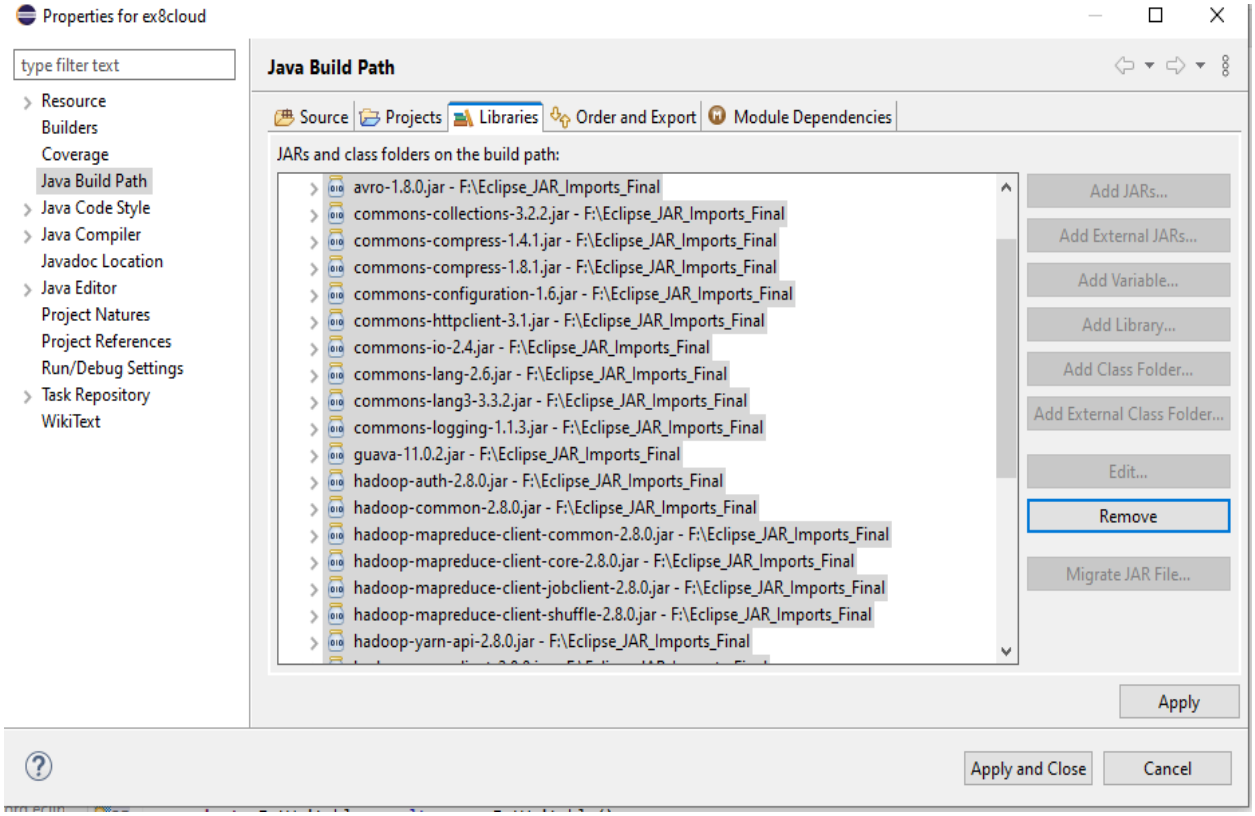
This PC
- 3D Objects
- Desktop
- Documents
- Downloads
- Music
- Pictures
- Videos
- Local Disk (C:)
- New Volume (F:)

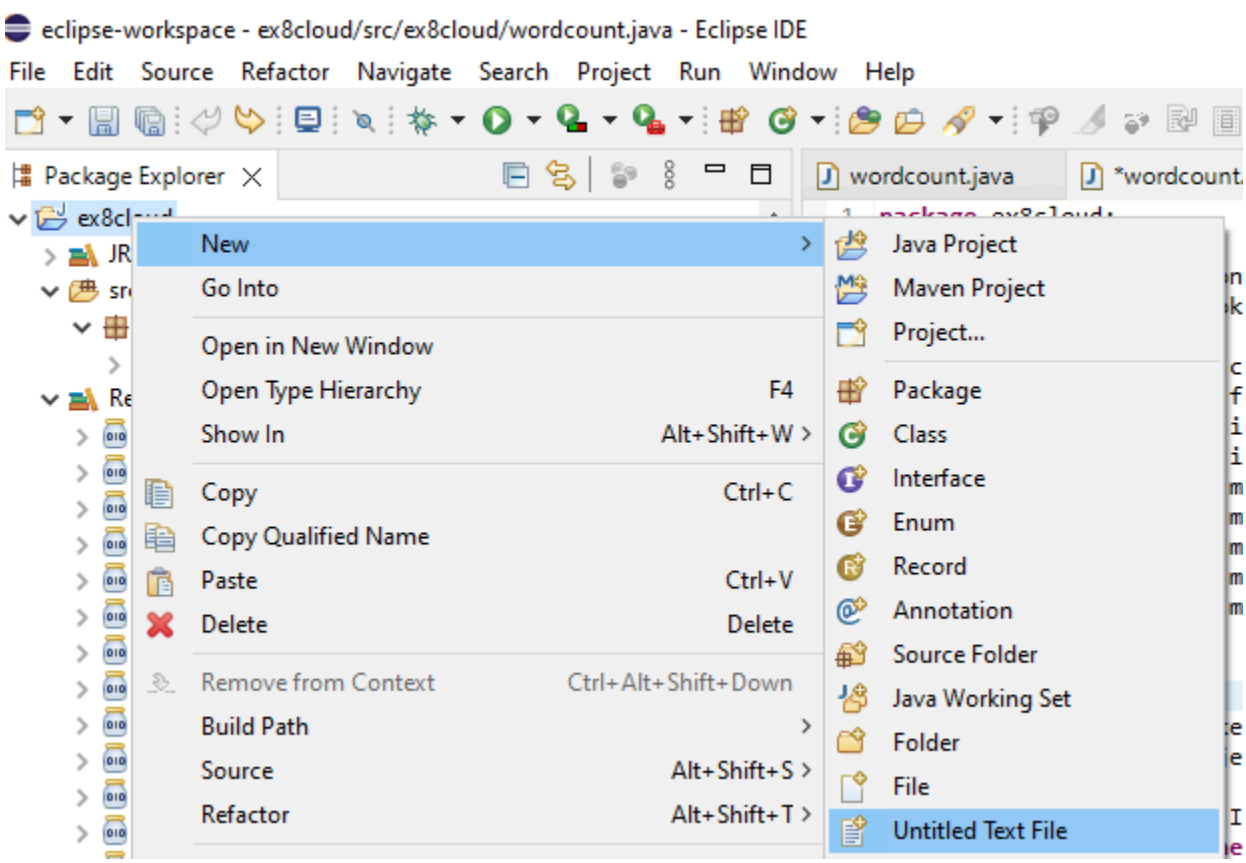
Network
- CE-LAB11
- DESKTOP-00IO8P6
- DESKTOP-3NN2NR9

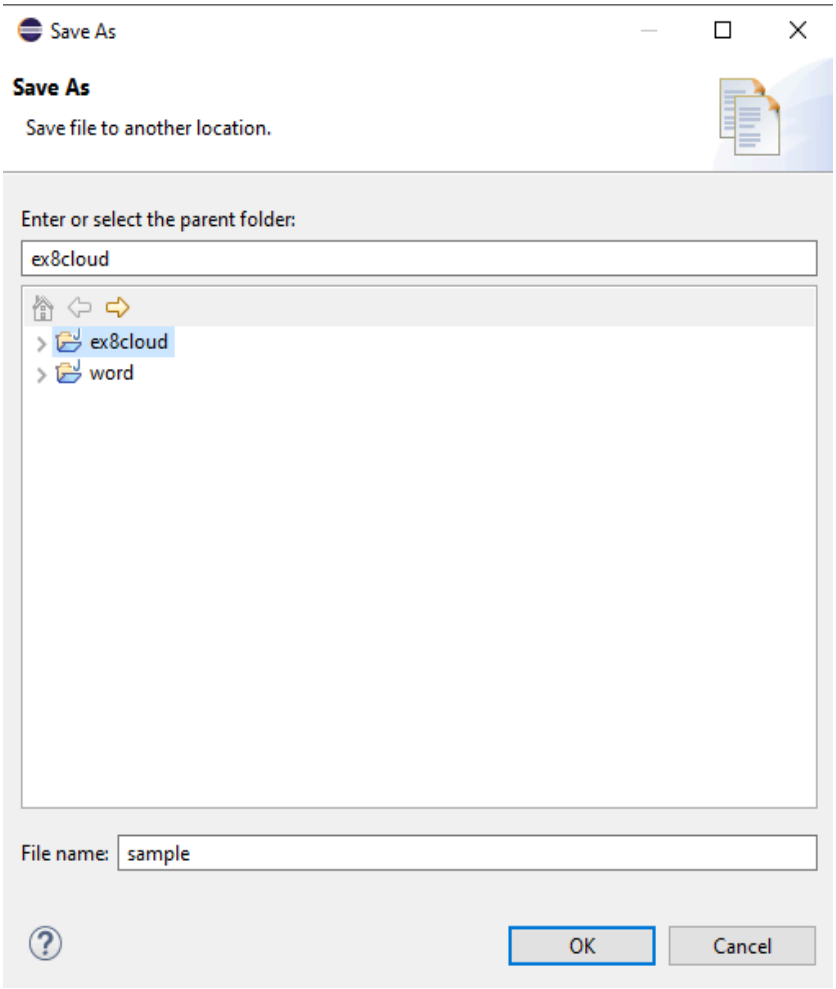
Name	Date modified	Type	Size
avro-1.8.0	07-05-2020 16:12	Executable Jar File	1,312 KB
commons-collections-3.2.2	17-03-2017 08:31	Executable Jar File	575 KB
commons-compress-1.4.1	17-03-2017 08:31	Executable Jar File	236 KB
commons-compress-1.8.1	07-05-2020 16:12	Executable Jar File	357 KB
commons-configuration-1.6	17-03-2017 08:31	Executable Jar File	292 KB
commons-httpclient-3.1	17-03-2017 08:31	Executable Jar File	298 KB
commons-io-2.4	17-03-2017 08:31	Executable Jar File	181 KB
commons-lang-2.6	17-03-2017 08:31	Executable Jar File	278 KB
commons-lang3-3.3.2	17-03-2017 08:31	Executable Jar File	404 KB
commons-logging-1.1.3	17-03-2017 08:31	Executable Jar File	61 KB
guava-11.0.2	17-03-2017 08:31	Executable Jar File	1,610 KB
hadoop-auth-2.8.0	17-03-2017 08:31	Executable Jar File	117 KB
hadoop-common-2.8.0	17-03-2017 11:11	Executable Jar File	3,647 KB
hadoop-mapreduce-client-common-2.8.0	17-03-2017 08:31	Executable Jar File	765 KB
hadoop-mapreduce-client-core-2.8.0	17-03-2017 08:31	Executable Jar File	1,535 KB
hadoop-mapreduce-client-jobclient-2.8.0	17-03-2017 08:31	Executable Jar File	66 KB
hadoop-mapreduce-client-shuffle-2.8.0	17-03-2017 08:31	Executable Jar File	74 KB
hadoop-yarn-api-2.8.0	17-03-2017 08:31	Executable Jar File	2,386 KB
hadoop-yarn-client-2.8.0	17-03-2017 08:31	Executable Jar File	225 KB
hadoop-yarn-common-2.8.0	17-03-2017 08:31	Executable Jar File	1,762 KB
htrace-core4-4.0.1-incubating	11-05-2020 20:41	Executable Jar File	1,451 KB
httpclient-4.5.2	11-05-2020 20:40	Executable Jar File	720 KB
httpcore	20-12-2012 04:35	Executable Jar File	169 KB
httpcore-4.4.1	11-05-2020 20:40	Executable Jar File	315 KB
jackson-core-asl-1.9.13	17-03-2017 08:31	Executable Jar File	227 KB
jackson-mapper-asl-1.9.13	11-05-2020 20:50	Executable Jar File	763 KB
log4j-1.2.17	07-05-2020 17:21	Executable Jar File	479 KB
log4j-api-2.5	07-05-2020 16:15	Executable Jar File	144 KB
log4j-core-2.5	07-05-2020 16:15	Executable Jar File	1,081 KB
slf4j-api-1.7.20	07-05-2020 16:18	Executable Jar File	40 KB

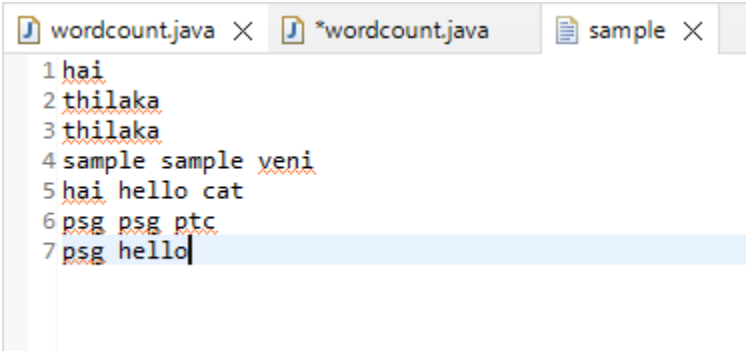

Properties for ex8cloud
type filter text
Resource
Builders
Coverage
Java Build Path
Java Code Style
Java Compiler
Javadoc Location
Java Editor
Project Natures
Project References
Run/Debug Settings
Task Repository
WikiText
Java Build Path
Source
Projects
Libraries
Order and Export
Module Dependencies
JARs and class folders on the build path:
Modulepath
JRE System Library [JavaSE-17]
Classpath
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit...
Remove
Migrate JAR File...
Apply
Apply and Close
Cancel

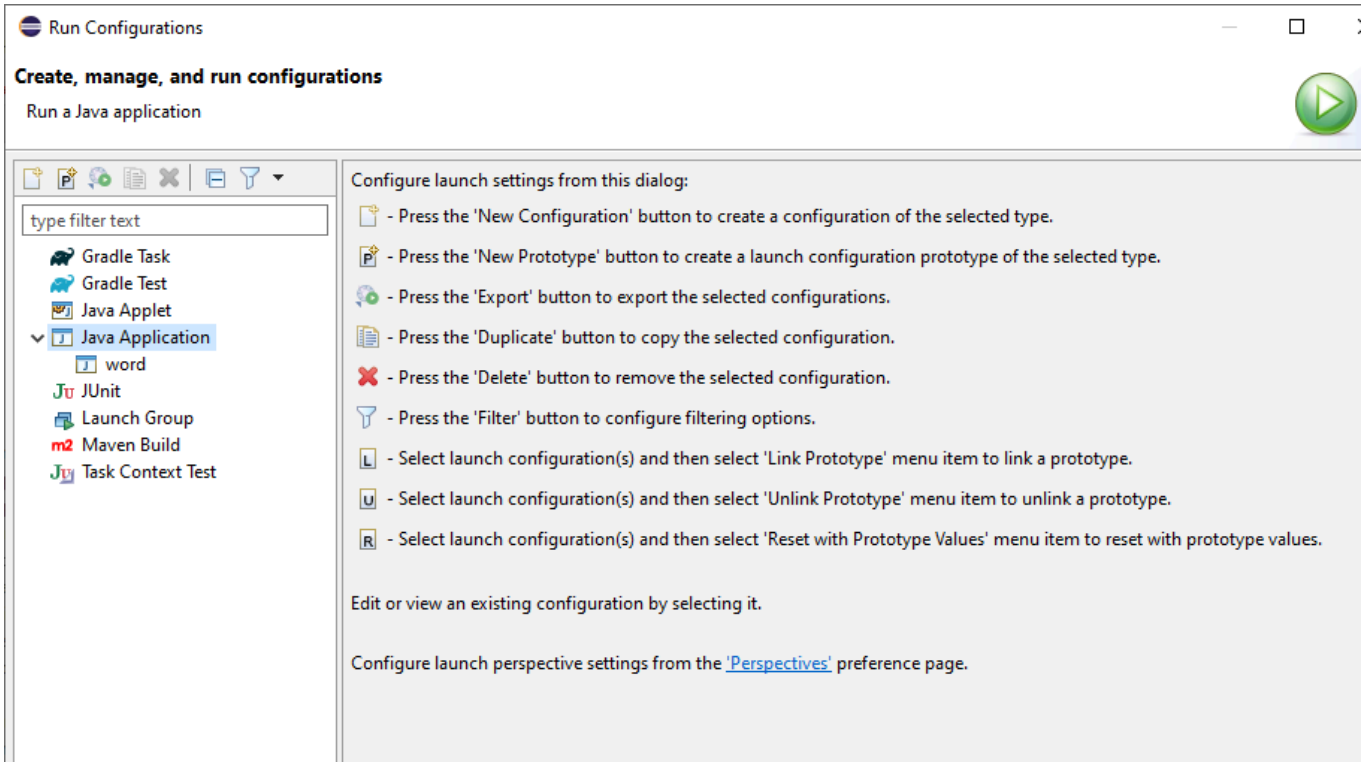

Properties for ex8cloud
JAR Selection
This PC › New Volume (F:) › Eclipse_JAR_Imports_Final
Search Eclipse_JAR_Imports_Fi...
Organize
New folder
New Volume (F:)
OneDrive
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
New Volume (F:)
Name
Date modified
Type
Size
avro-1.8.0 07-05-2020 16:12 Executable Jar File 1,312 KB
commons-collections-3.2.2 17-03-2017 08:31 Executable Jar File 575 KB
commons-compress-1.4.1 17-03-2017 08:31 Executable Jar File 236 KB
commons-compress-1.8.1 07-05-2020 16:12 Executable Jar File 357 KB
commons-configuration-1.6 17-03-2017 08:31 Executable Jar File 292 KB
commons-httpclient-3.1 17-03-2017 08:31 Executable Jar File 298 KB
commons-io-2.4 17-03-2017 08:31 Executable Jar File 181 KB
commons-lang-2.6 17-03-2017 08:31 Executable Jar File 278 KB
commons-lang3-3.3.2 17-03-2017 08:31 Executable Jar File 404 KB
commons-logging-1.1.3 17-03-2017 08:31 Executable Jar File 61 KB
guava-11.0.2 17-03-2017 08:31 Executable Jar File 1,610 KB
hadoop-auth-2.8.0 17-03-2017 08:31 Executable Jar File 117 KB
File name: "commons-compress-1.4.1" "commons-compress-1.8.1" "commor
.jar;.zip
Open
Cancel

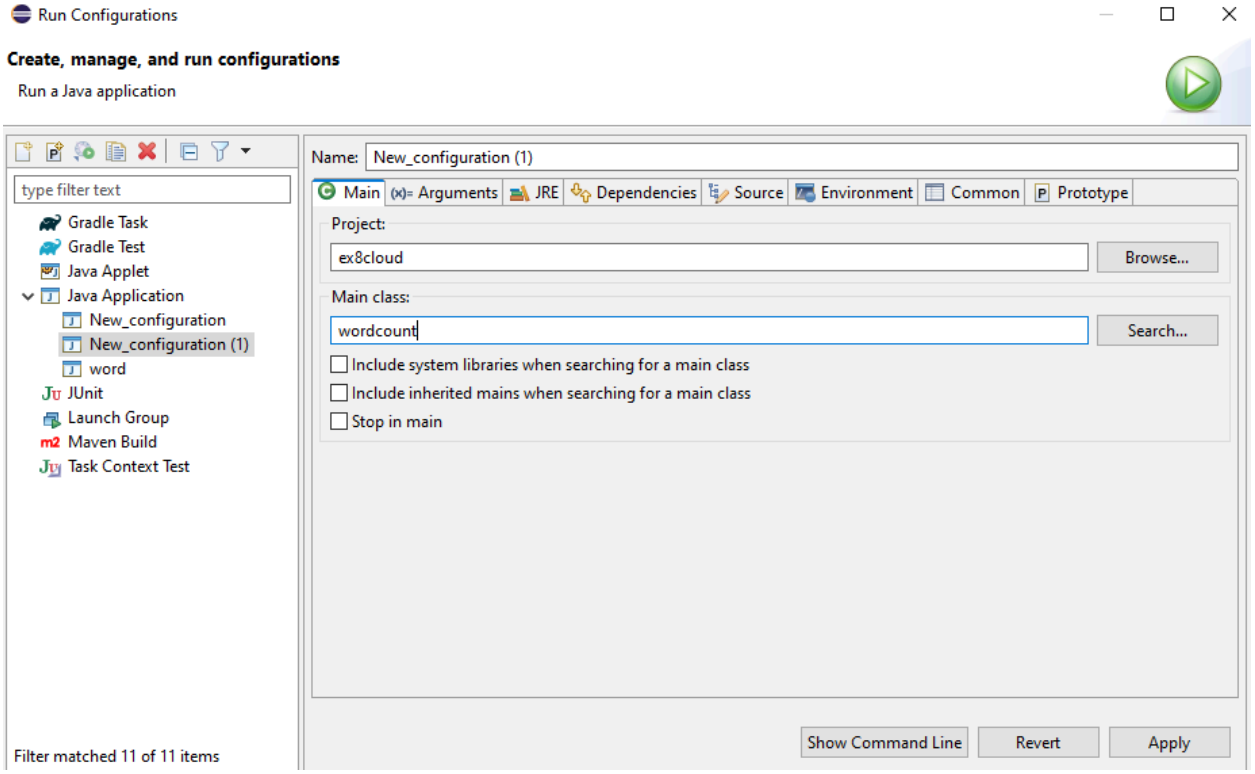

Properties for ex8cloud
type filter text
Resource
Builders
Coverage
Java Build Path
Java Code Style
Java Compiler
Javadoc Location
Java Editor
Project Natures
Project References
Run/Debug Settings
Task Repository
WikiText
Java Build Path
Source
Projects
Libraries
Order and Export
Module Dependencies
JARs and class folders on the build path:
avro-1.8.0.jar - F:\Eclipse_JAR_Imports_Final
commons-collections-3.2.2.jar - F:\Eclipse_JAR_Imports_Final
commons-compress-1.4.1.jar - F:\Eclipse_JAR_Imports_Final
commons-compress-1.8.1.jar - F:\Eclipse_JAR_Imports_Final
commons-configuration-1.6.jar - F:\Eclipse_JAR_Imports_Final
commons-httpclient-3.1.jar - F:\Eclipse_JAR_Imports_Final
commons-io-2.4.jar - F:\Eclipse_JAR_Imports_Final
commons-lang-2.6.jar - F:\Eclipse_JAR_Imports_Final
commons-lang3-3.3.2.jar - F:\Eclipse_JAR_Imports_Final
commons-logging-1.1.3.jar - F:\Eclipse_JAR_Imports_Final
guava-11.0.2.jar - F:\Eclipse_JAR_Imports_Final
hadoop-auth-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-common-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-mapreduce-client-common-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-mapreduce-client-core-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-mapreduce-client-jobclient-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-mapreduce-client-shuffle-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
hadoop-yarn-api-2.8.0.jar - F:\Eclipse_JAR_Imports_Final
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit...
Remove
Migrate JAR File...
Apply
Apply and Close
Cancel


eclipse-workspace - ex8cloud/src/ex8cloud/wordcount.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
wordcount.java
*wordcount.
New
Go Into
Open in New Window
Open Type Hierarchy F4
Show In Alt+Shift+W
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Remove from Context Ctrl+Alt+Shift+Down
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Java Project
Maven Project
Project...
Package
Class
Interface
Enum
Record
Annotation
Source Folder
Java Working Set
Folder
File
Untitled Text File


Save As
Save As
Save file to another location.
Enter or select the parent folder:
ex8cloud
ex8cloud
word
File name:
sample
OK
Cancel


wordcount.java
*wordcount.java
sample
1 hai
2 thilaka
3 thilaka
4 sample sample veni
5 hai hello cat
6 psg psg ptc
7 psg hello

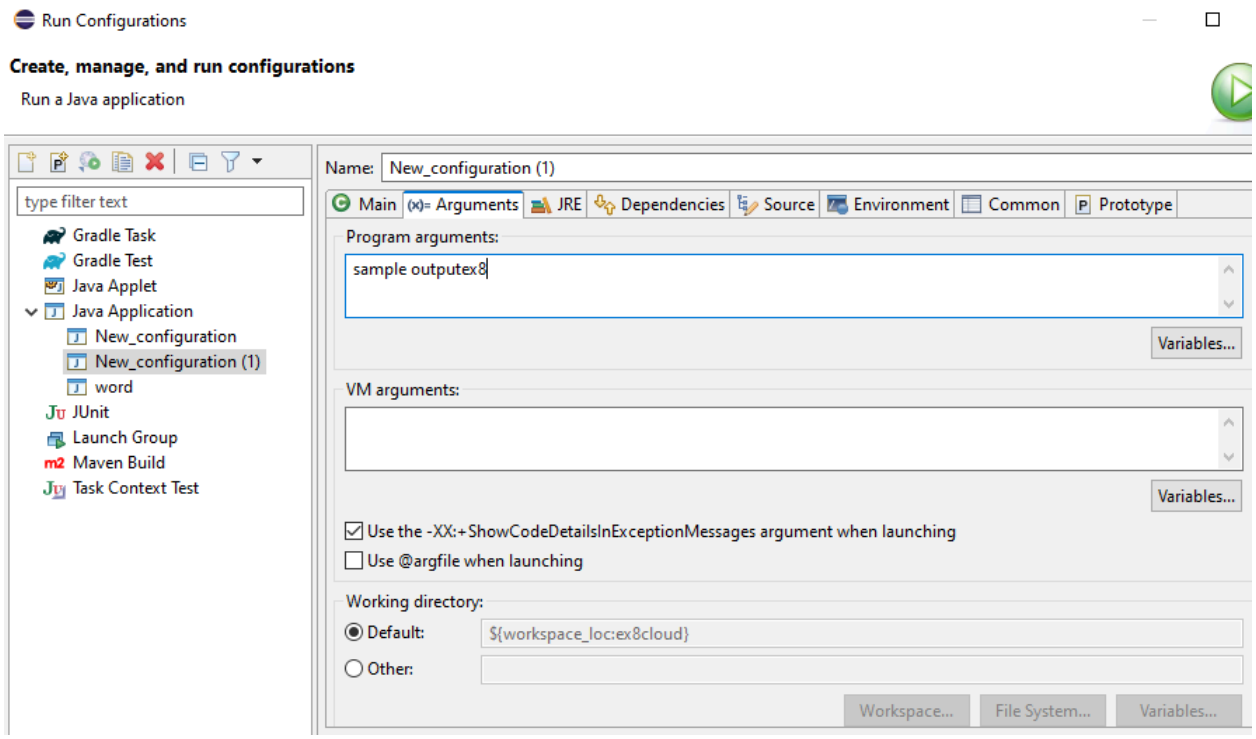

Run Configurations
Create, manage, and run configurations
Run a Java application
type filter text
Gradle Task
Gradle Test
Java Applet
Java Application
word
JUnit
Launch Group
Maven Build
Task Context Test
Configure launch settings from this dialog:
- Press the 'New Configuration' button to create a configuration of the selected type.
- Press the 'New Prototype' button to create a launch configuration prototype of the selected type.
- Press the 'Export' button to export the selected configurations.
- Press the 'Duplicate' button to copy the selected configuration.
- Press the 'Delete' button to remove the selected configuration.
- Press the 'Filter' button to configure filtering options.
- Select launch configuration(s) and then select 'Link Prototype' menu item to link a prototype.
- Select launch configuration(s) and then select 'Unlink Prototype' menu item to unlink a prototype.
- Select launch configuration(s) and then select 'Reset with Prototype Values' menu item to reset with prototype values.
Edit or view an existing configuration by selecting it.
Configure launch perspective settings from the 'Perspectives' preference page.

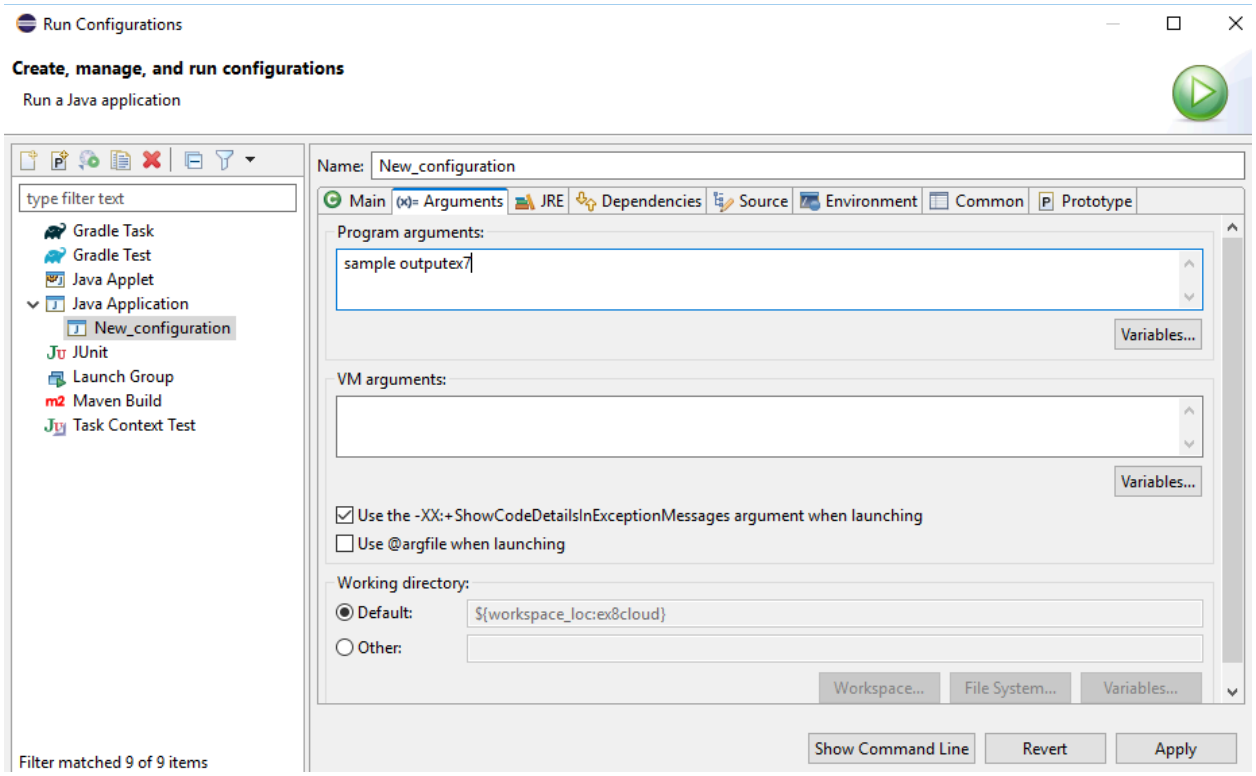

Run Configurations
Create, manage, and run configurations
Run a Java application
type filter text
Gradle Task
Gradle Test
Java Applet
Java Application
New_configuration
New_configuration (1)
word
JUnit
Launch Group
Maven Build
Task Context Test
Filter matched 11 of 11 items
Name: New_configuration (1)
Main
Arguments
JRE
Dependencies
Source
Environment
Common
Prototype
Project:
ex8cloud
Browse...
Main class:
wordcount
Search...
Include system libraries when searching for a main class
Include inherited mains when searching for a main class
Stop in main
Show Command Line
Revert
Apply

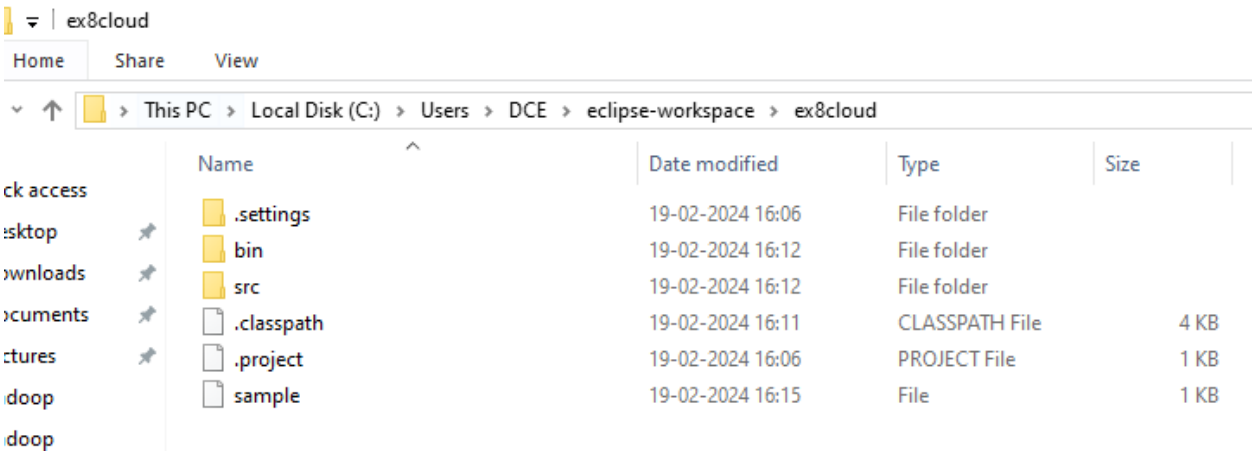
part-r-00000 - Notepad

File Edit Format View Help

```
cat     1
hai     2
hello   2
psg     3
ptc     1
sample  2
thilaka 2
veni    1
```




Run Configurations
Create, manage, and run configurations
Run a Java application
type filter text
Gradle Task
Gradle Test
Java Applet
Java Application
New_configuration
JUnit
Launch Group
Maven Build
Task Context Test
Filter matched 9 of 9 items
Name: New_configuration
Main
Arguments
JRE
Dependencies
Source
Environment
Common
Prototype
Program arguments:
sample outputex7
Variables...
VM arguments:
Variables...
Use the -XX:+ShowCodeDetailsInExceptionMessages argument when launching
Use @argfile when launching
Working directory:
Default:
${workspace_loc:ex8cloud}
Other:
Workspace...
File System...
Variables...
Show Command Line
Revert
Apply


ex8cloud
Home
Share
View
This PC › Local Disk (C:) › Users › DCE › eclipse-workspace › ex8cloud
Name
Date modified
Type
Size
.settings
19-02-2024 16:06
File folder
bin
19-02-2024 16:12
File folder
src
19-02-2024 16:12
File folder
.classpath
19-02-2024 16:11
CLASSPATH File
4 KB
.project
19-02-2024 16:06
PROJECT File
1 KB
sample
19-02-2024 16:15
File
1 KB

ex8cloud

Share View

This PC > Local Disk (C:) > Users > admin > eclipse-workspace > ex8cloud

Name	Date modified	Type
.settings	2/19/2024 8:53 PM	File folder
bin	2/19/2024 8:59 PM	File folder
outputex7	2/19/2024 9:00 PM	File folder
src	2/19/2024 8:54 PM	File folder
.classpath	2/19/2024 8:55 PM	CLASSPATH File
.project	2/19/2024 8:53 PM	PROJECT File
sample	2/19/2024 8:57 PM	File

outputex7

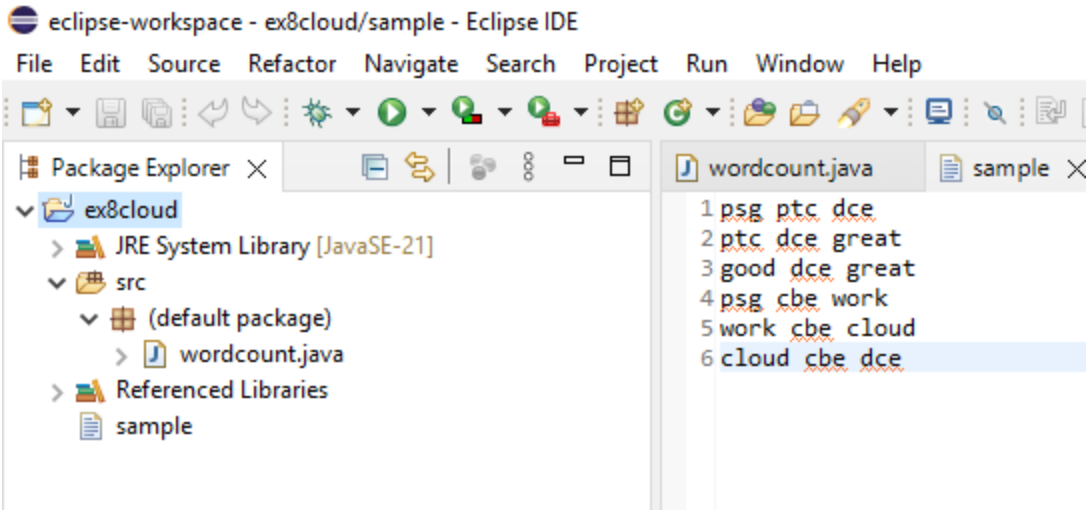
Share View

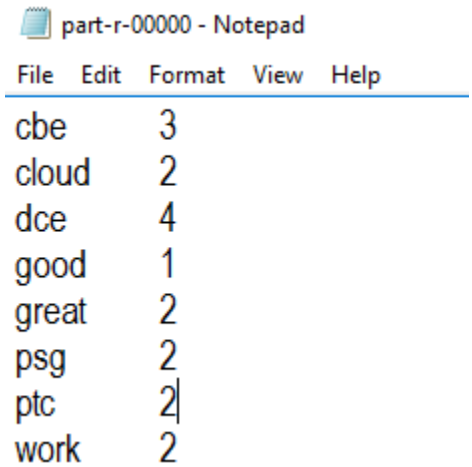
This PC > Local Disk (C:) > Users > admin > eclipse-workspace > ex8cloud > outputex7

Name	Date modified	Type
._SUCCESS.crc	2/19/2024 9:00 PM	CRC File
.part-r-00000.crc	2/19/2024 9:00 PM	CRC File
_SUCCESS	2/19/2024 9:00 PM	File
part-r-00000	2/19/2024 9:00 PM	File

wordcount.java | sample

```
psg ptc dce
ptc dce great
good dce great
psg cbe work
work cbe cloud
cloud cbe dce
```





Result

Thus the application like word count was created using hadoop.